

Unbiased estimation of normalized scale-invariant indices under the Gamma distribution: Applications to GDP per capita in the Americas

1 Introduction

This document provides the codes for estimation and one data fitting using the unbiased estimation of **normalized scale-invariant indices** (NSII) under the Gamma distribution¹ (cf. Vila et al., 2026). For more details on theoretical properties of the estimator see <https://doi.org/10.48550/arXiv.2606.22712>.

2 Data

In order to show the versatility of the {NSII} estimator, we show that the framework successfully fits the distribution of Gross Domestic Product (GDP) per capita across the Americas. The dataset includes $n = 34$ countries and territories in the Americas in 2023, measured in international dollars at 2021 PPP prices. The data were obtained from <https://ourworldindata.org/grapher/gdp-per-capita-worldbank> (accessed on June 2026). Also available in `Data/data.csv`.

```
countries <- c(
  "Haiti", "Honduras", "Nicaragua", "Bolivia", "Jamaica",
  "El Salvador", "Guatemala", "Belize", "Ecuador", "Peru",
  "Paraguay", "Grenada", "Dominica", "St. Vincent and the Grenadines",
  "Colombia", "Brazil", "Suriname", "Barbados", "Mexico",
  "Dominican Republic", "Saint Lucia", "Costa Rica", "Argentina",
  "Antigua and Barbuda", "Chile", "St. Kitts and Nevis",
  "Uruguay", "Trinidad and Tobago", "Bahamas", "Panama",
  "Puerto Rico", "Guyana", "Canada", "United States")

gdp <- c(
  2.956, 6.468, 7.487, 9.844, 10.291, 11.404, 12.389, 12.455,
  14.472, 15.294, 15.783, 16.946, 17.420, 18.335, 18.692, 19.018,
  19.044, 19.224, 22.143, 23.088, 23.403, 26.293, 27.105, 28.967,
  29.463, 30.409, 31.019, 31.706, 33.106, 35.864, 42.995, 49.315,
  55.919, 74.578)
names(gdp) <- countries
n <- length(gdp)
summary(gdp)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.96   14.68   19.13   23.91   30.17   74.58
```

3 Geographic heatmap

```
#required packages: sf, ggplot2, rnatuarearth, dplyr
library(sf)
library(ggplot2)
library(rnatuarearth)
library(dplyr)
```

```

# Data
dados <- data.frame(
  country = countries,
  gdp = gdp
)

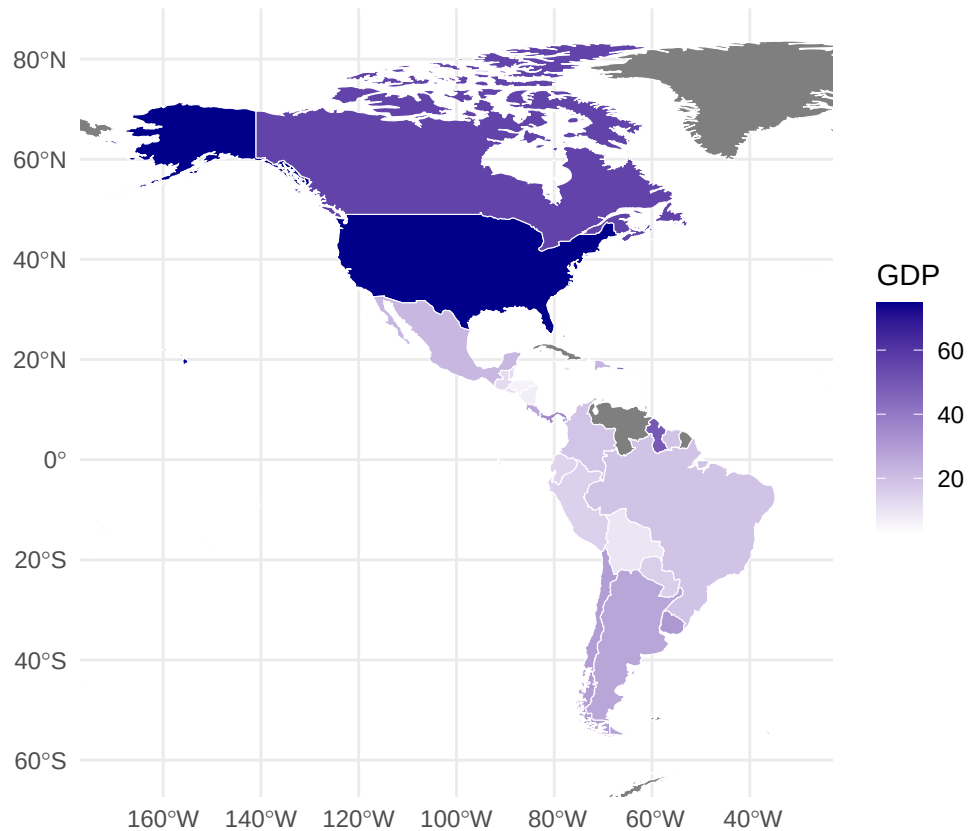
# Mapa mundial
world <- ne_countries(scale = "medium", returnclass = "sf")

# Ajustes de nomes para coincidir com o mapa
dados$country[dados$country == "United States"] <- "United States of America"
dados$country[dados$country == "St. Vincent & Gren."] <- "Saint Vincent and the Grenadines"
dados$country[dados$country == "Dominican Republic"] <- "Dominican Rep."
dados$country[dados$country == "St. Kitts and Nevis"] <- "Saint Kitts and Nevis"

mapa <- world %>%
  left_join(dados, by = c("name" = "country"))

#x11()
# Heatmap
ggplot(mapa) +
  geom_sf(aes(fill = gdp), color = "white", linewidth = 0.2) +
  scale_fill_gradient(
    low = "white",
    high = "darkblue",
    name = "GDP"
  ) +
  coord_sf(
    xlim = c(-170, -30),
    ylim = c(-60, 85)
  ) +
  # labs(
  #   title = "GDP per capita nas Américas",
  #   subtitle = "Países da amostra"
  # ) +
  theme_minimal()

```



4 Gamma fit

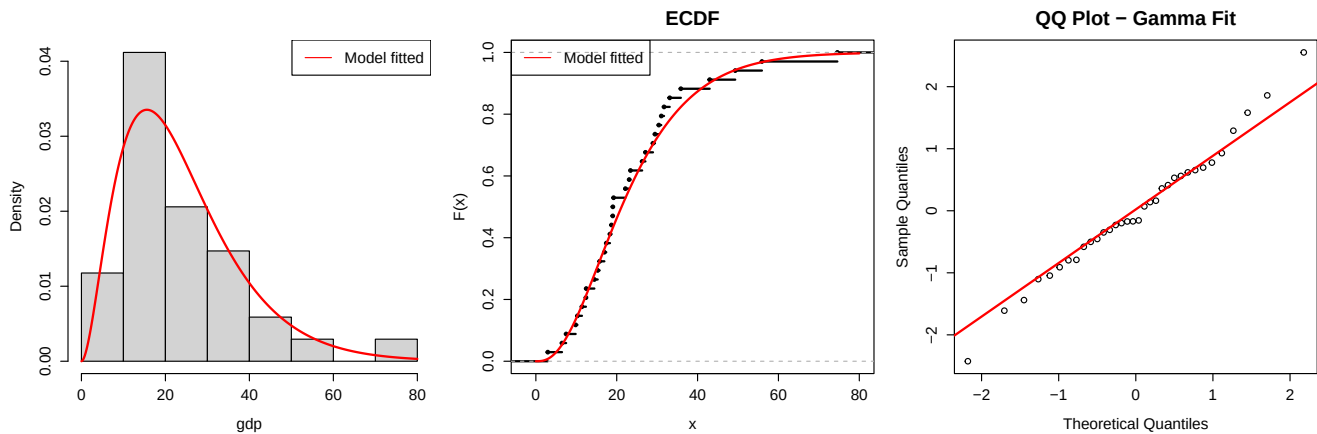
```
gamma_mle <- function(x) {
  s <- log(mean(x)) - mean(log(x))
  f <- function(a) log(a) - digamma(a) - s
  lo <- 1e-6; hi <- 1
  while (f(hi) > 0) hi <- hi * 2
  while (f(lo) < 0) lo <- lo / 2
  a <- uniroot(f, c(lo, hi), tol = .Machine$double.eps^0.75)$root
  c(alpha = a, lambda = a / mean(x))
}

pars <- gamma_mle(gdp)
al <- pars[["alpha"]]; la <- pars[["lambda"]]
pars

## alpha lambda
## 2.8761 0.1203

par(mfrow = c(1, 3), mar = c(4.5, 4.5, 3, 1))
xs <- seq(0, 80, length.out = 400)
hist(gdp, probability = TRUE, breaks = 10, main = "", xlab = "gdp",
     cex.axis = 1.3, cex.lab = 1.3)
lines(xs, dgamma(xs, al, la), col = "red", lwd = 2)
legend("topright", "Model fitted", lty = 1, col = "red", cex = 1.2)
plot(ecdf(gdp), main = "ECDF", xlab = "x", ylab = "F(x)", lwd = 2,
     cex.main = 1.6, cex.axis = 1.3, cex.lab = 1.3)
lines(xs, pgamma(xs, al, la), col = "red", lwd = 2)
legend("topleft", "Model fitted", lty = 1, col = "red", cex = 1.2)
```

```
res <- qnorm(pgamma(sort(gdp), al, la))
qqnorm(res, main = "QQ Plot - Gamma Fit", cex.main = 1.6,
        cex.axis = 1.3, cex.lab = 1.3)
qqline(res, col = "red", lwd = 2)
```



5 Goodness of fit

Parametric bootstrap p-values ($B = 10,000$), both parameters re-estimated in each sample.

```
gof_stats <- function(x, alpha, lambda) {
  u <- sort(pgamma(x, shape = alpha, rate = lambda))
  n <- length(u); i <- seq_len(n)
  c(KS = max(pmax(i/n - u, u - (i - 1)/n)),
    CvM = sum((u - (2*i - 1)/(2*n))^2) + 1/(12*n),
    AD = -n - mean((2*i - 1) * (log(u) + log1p(-rev(u)))))
}
obs_gof <- gof_stats(gdp, al, la)
set.seed(SEED)
boot_gof <- replicate(B_GOF, {
  xb <- rgamma(n, shape = al, rate = la)
  pb <- gamma_mle(xb)
  gof_stats(xb, pb[["alpha"]], pb[["lambda"]])
})
gof_tab <- data.frame(Test = c("Kolmogorov-Smirnov", "Cramer-von Mises",
                              "Anderson-Darling"),
                     Statistic = obs_gof,
                     `p-value` = rowMeans(boot_gof >= obs_gof),
                     row.names = NULL, check.names = FALSE)
knitr::kable(gof_tab, digits = 4, booktabs = TRUE)
```

Test	Statistic	p-value
Kolmogorov-Smirnov	0.0913	0.6808
Cramer-von Mises	0.0294	0.8670
Anderson-Darling	0.1936	0.9066

6 Estimators

```
# m = 2 kernels and sharp constants C_g
p <- 0.5; th <- 1; bt <- 1
```

```

K2 <- list(
  Gini      = list(g = function(a, b) abs(a - b), Cg = 1),
  Power     = list(g = function(a, b) (a^p + b^p) * (a + b)^(1 - p),
                    Cg = max(1, 2^(1 - p))),
  Symmetric = list(g = function(a, b) a^th * b^bt / (a + b)^(th + bt - 1),
                    Cg = th^th * bt^bt / (th + bt)^(th + bt)),
  Maximum   = list(g = function(a, b) pmax(a, b), Cg = 1),
  Minimum   = list(g = function(a, b) pmin(a, b), Cg = 0.5),
  SCV       = list(g = function(a, b) (a - b)^2 / (a + b), Cg = 1),
  Product   = list(g = function(a, b) a * b / (a + b), Cg = 1/4)
)

# Order-statistic kernels:  $\sum_{|S|=m} x_{[k:S]} = \sum_i C(i-1, k-1) C(n-i, m-k) x_{(i)}$ 
os_coef <- function(n, m, a) {
  i <- seq_len(n); cc <- numeric(n)
  for (k in seq_len(m))
    if (a[k] != 0) cc <- cc + a[k] * choose(i - 1, k - 1) * choose(n - i, m - k)
  cc / choose(n, m)
}

hat_I      <- function(x, m, a, Cg) sum(os_coef(length(x), m, a) * sort(x)) /
  (Cg * m * mean(x))
hat_I_mat <- function(Xs, m, a, Cg, xbar)
  as.vector(crossprod(os_coef(nrow(Xs), m, a), Xs)) / (Cg * m * xbar)

# Theoretical indices under Gamma(alpha, 1)
aux_A <- function(alpha, r)
  integrate(function(t) 1 - pgamma(t, shape = alpha)^r, 0, Inf,
    rel.tol = 1e-11, subdivisions = 500L)$value
EX <- function(alpha, m, k) {
  r <- k:m
  sum((-1)^(r - k) * choose(r - 1, k - 1) * choose(m, r) *
    vapply(r, function(rr) aux_A(alpha, rr), numeric(1)))
}
I_lin <- function(alpha, a) {
  m <- length(a)
  sum(a * vapply(seq_len(m), function(k) EX(alpha, m, k), numeric(1))) / (m * alpha)
}

cfg <- function(family, a, Cg = 1, conf = "--")
  list(family = family, m = length(a), a = a, Cg = Cg, conf = conf,
    Ifun = function(alpha) I_lin(alpha, a) / Cg)
ext <- function(m, j, k) {
  a <- numeric(m); a[j] <- -1; a[k] <- 1
  cfg("Extended mth Gini", a, 1/(m - k + 1), sprintf("(%d,%d)", j, k))
}
TAB6 <- list(
  cfg("mth Gini", c(-1, 0, 1)), cfg("mth Gini", c(-1, 0, 0, 1)),
  ext(3, 1, 2), ext(3, 2, 3), ext(4, 1, 2), ext(4, 2, 3), ext(4, 3, 4),
  cfg("Linear Order-Statistic", c(-1, 0, 0, 1), conf = "(-1,0,0,1)"),
  cfg("Linear Order-Statistic", c(-1, 0, 0, 0, 1), conf = "(-1,0,0,0,1)")
)

```

7 Bootstrap

Non-parametric: resample countries with replacement. Parametric: draw from $\text{Gamma}(\hat{\alpha}, \hat{\lambda})$ and re-estimate both parameters. Percentile intervals, $B = 5,000$.

```
qs <- c((1 - CONF)/2, 1 - (1 - CONF)/2)
set.seed(SEED)
XB <- matrix(sample(gdp, n * B_BOOT, replace = TRUE), n, B_BOOT)
XBs <- apply(XB, 2, sort)
xbarB <- colMeans(XB)
alpha_b <- vapply(seq_len(B_BOOT), function(b)
  gamma_mle(rgamma(n, shape = al, rate = la))["alpha"], numeric(1))

make_spline <- function(lfun, n_grid = 400L) {
  gr <- seq(max(0.2, min(alpha_b) * 0.98), max(alpha_b) * 1.02, length.out = n_grid)
  splinefun(gr, vapply(gr, lfun, numeric(1)), method = "natural")
}
fmt_ci <- function(lo, hi) sprintf("%.4f, %.4f", lo, hi)
```

7.1 Indices with $m = 2$

```
prs <- combn(n, 2L)
Ai <- XB[prs[1L, ], , drop = FALSE]
Aj <- XB[prs[2L, ], , drop = FALSE]

tab5 <- do.call(rbind, lapply(names(K2), function(nm) {
  k <- K2[[nm]]
  bval <- colMeans(k$g(Ai, Aj)) / (k$Cg * 2 * xbarB)
  ci <- quantile(bval, qs, names = FALSE)
  data.frame(Index = nm, Cg = k$Cg,
    Estimate = mean(k$g(gdp[prs[1L, ]], gdp[prs[2L, ]])) /
      (k$Cg * 2 * mean(gdp)),
    SE = sd(bval), `95% CI` = fmt_ci(ci[1], ci[2]), check.names = FALSE)
}))
knitr::kable(tab5, digits = 4, booktabs = TRUE, linesep = "",
  col.names = c("Index", "$C_g$", "$\\widehat{I}_{g,2}$", "SE", "95\\% CI"),
  escape = FALSE)
```

Index	C_g	$\widehat{I}_{g,2}$	SE	95% CI
Gini	1.000	0.3286	0.0397	[0.2407, 0.3951]
Power	1.414	0.9777	0.0053	[0.9674, 0.9878]
Symmetric	0.250	0.8456	0.0327	[0.7835, 0.9107]
Maximum	1.000	0.6643	0.0199	[0.6203, 0.6975]
Minimum	0.500	0.6714	0.0397	[0.6049, 0.7593]
SCV	1.000	0.1544	0.0327	[0.0893, 0.2165]
Product	0.250	0.8456	0.0327	[0.7835, 0.9107]

7.2 Order-statistic indices

```
tab6 <- do.call(rbind, lapply(TAB6, function(cf) {
  b_np <- hat_I_mat(XBs, cf$m, cf$a, cf$Cg, xbarB)
  b_pr <- make_spline(cf$Ifun)(alpha_b)
  ci_np <- quantile(b_np, qs, names = FALSE)
  ci_pr <- quantile(b_pr, qs, names = FALSE)
  data.frame(Index = cf$family, m = cf$m, Config = cf$conf,
```

```

    np_est = hat_I(gdp, cf$m, cf$a, cf$Cg), np_se = sd(b_np),
    np_ci = fmt_ci(ci_np[1], ci_np[2]),
    par_est = cf$Ifun(al), par_se = sd(b_pr),
    par_ci = fmt_ci(ci_pr[1], ci_pr[2]))
  )))
k6 <- knitr::kable(tab6, format = "latex", digits = 4, booktabs = TRUE, linesep = "",
  escape = FALSE,
  col.names = c("Index", "$m$", "Config.",
    "$\\widehat{I}_{g,m}$", "SE", "95\\% CI",
    "$I_m(\\hat{\\alpha})$", "SE", "95\\% CI"))
knitr::asis_output(paste0("\\begin{group}\\small", k6, "\\end{group}"))

```

Index	m	Config.	$\widehat{I}_{g,m}$	SE	95% CI	$I_m(\hat{\alpha})$	SE	95% CI
nth Gini	3	—	0.3286	0.0397	[0.2407, 0.3951]	0.3186	0.0336	[0.2451, 0.3776]
nth Gini	4	—	0.3029	0.0367	[0.2200, 0.3624]	0.2910	0.0308	[0.2238, 0.3452]
Extended nth Gini	3	(1,2)	0.2443	0.0286	[0.1845, 0.2974]	0.2565	0.0201	[0.2092, 0.2883]
Extended nth Gini	3	(2,3)	0.2064	0.0334	[0.1320, 0.2604]	0.1903	0.0236	[0.1405, 0.2335]
Extended nth Gini	4	(1,2)	0.2125	0.0291	[0.1510, 0.2654]	0.2195	0.0133	[0.1858, 0.2381]
Extended nth Gini	4	(2,3)	0.1540	0.0222	[0.1132, 0.2009]	0.1653	0.0168	[0.1280, 0.1944]
Extended nth Gini	4	(3,4)	0.1551	0.0292	[0.0880, 0.2006]	0.1352	0.0180	[0.0978, 0.1687]
Linear Order-Statistic	4	(-1,0,0,1)	0.3029	0.0367	[0.2200, 0.3624]	0.2910	0.0308	[0.2238, 0.3452]
Linear Order-Statistic	5	(-1,0,0,0,1)	0.2773	0.0339	[0.1996, 0.3310]	0.2635	0.0280	[0.2024, 0.3128]

Columns 4–6: non-parametric bootstrap; columns 7–9: parametric bootstrap.

R 4.5.1